Rental Management

A rental system for small landlords, with a tenant portal

Lidor Amraby · final project · rental-management-app-wine.vercel.app

What it is

- Buildings and the units inside them
- Tenancies: who rents what, for how long, at what rent
- A rent ledger the landlord writes as money arrives
- Repairs, reported by the tenant and followed to the end

It records money received. It is not a payment processor.

The problem

A spreadsheet, a phone, and a memory.

- What is owed is arithmetic somebody does by hand
- The tenant cannot check anything without asking
- A repair promised by phone leaves no record

The users

- **Landlord** — owns the portfolio, records everything
- **Tenant** — reads their own tenancy and its ledger
- A tenant has two writes: report a problem, confirm a repair

Neither of them is rent. Only the landlord records money.

The business value

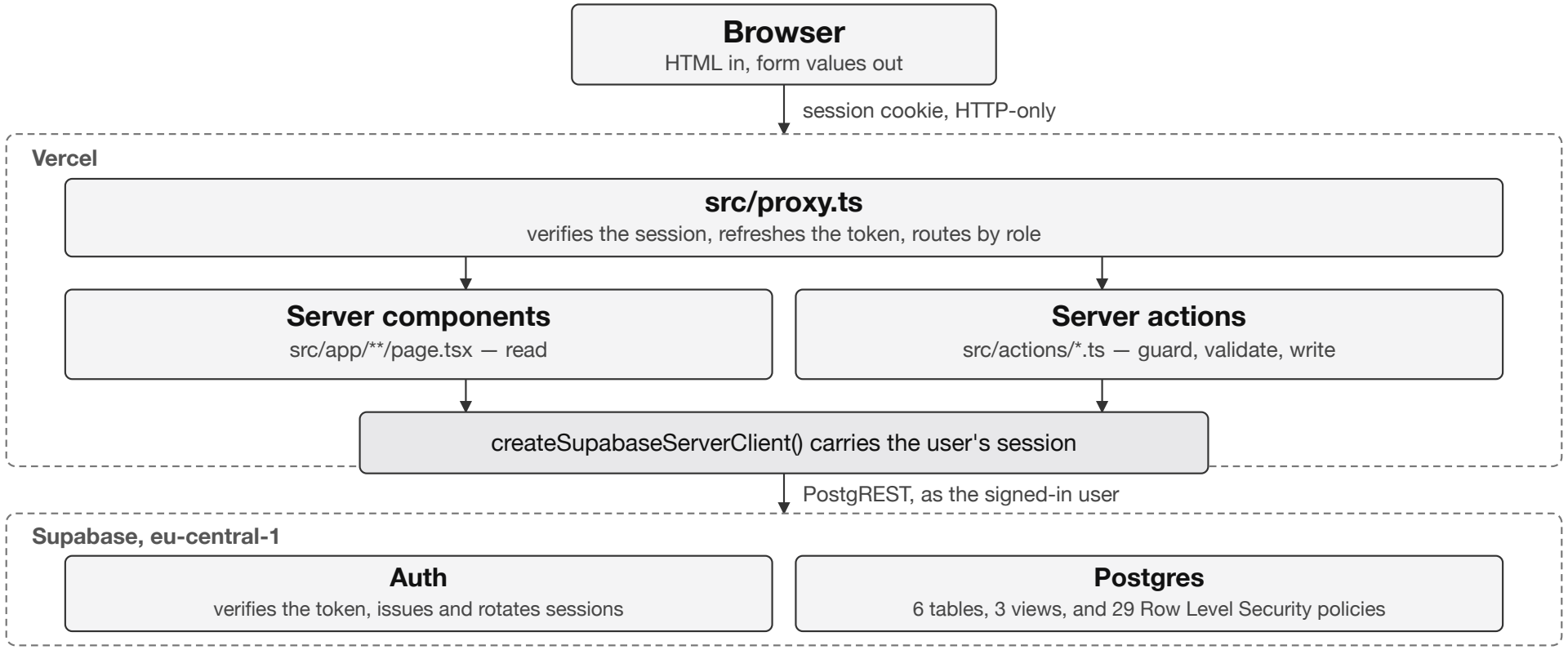
- Arrears visible at a glance, always current
- Fewer phone calls: the tenant sees their own position
- Every repair has a timestamped route and a confirmation

How it is built

- Next.js App Router, TypeScript, deployed on Vercel
- Supabase: Postgres, Auth, Row Level Security
- Server components read. Server actions write.
- No database client in the browser at all

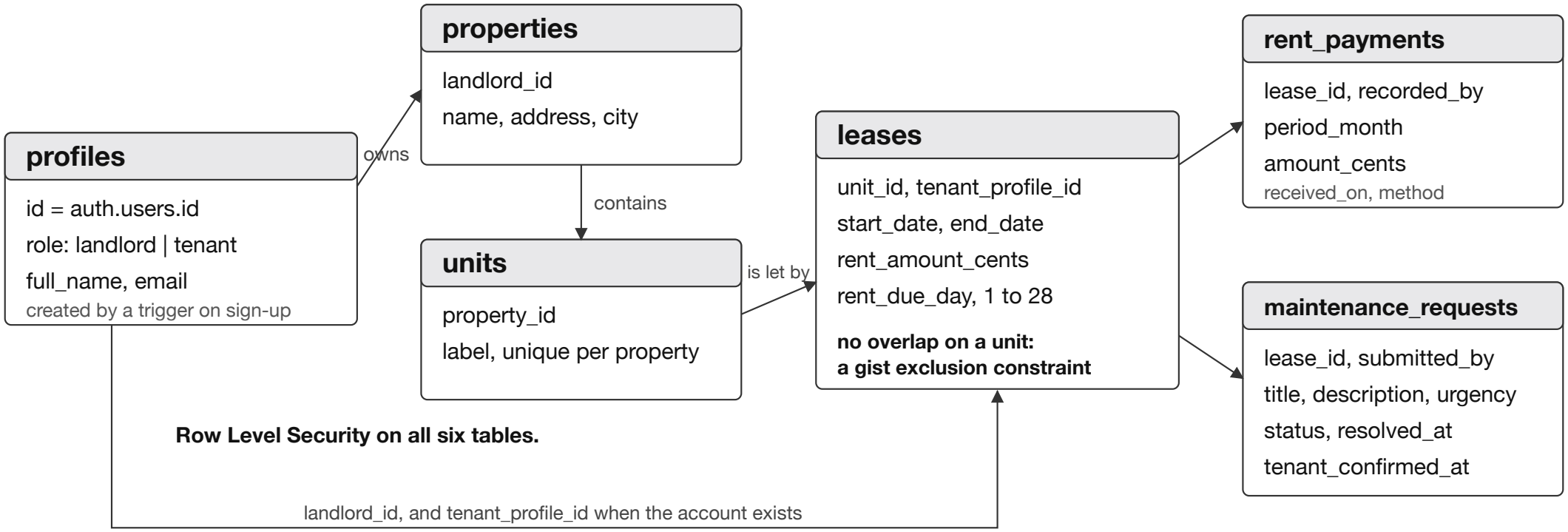
Authorisation lives in the database, not in the application.

The architecture



The routing is a convenience. The database is the boundary that refuses.

The database



No lease status column, no occupancy flag, no rent periods table. All derived.

Every write, the same seven steps

1. Resolve the acting user from the session
2. Refuse anyone who is not the right role
3. Parse the input with the form's own schema
4. Apply the business rules that need other rows
5. Write, with Row Level Security as the last word
6. Revalidate the pages that changed
7. Return a typed result the form can render

The landlord id comes from the session, never from the form.

Demo

rental-management-app-wine.vercel.app

Including two deliberate failures: an overlapping tenancy, and one tenant reaching for another's data

The tests

- 346 unit and component tests — the rules at their boundaries
- 134 permission and database tests — against a real Postgres
- 22 end-to-end tests — whole processes in a browser
- 5 documented manual checks — print, layout, screen reader

The permission tests attack the database, not the interface.

Scale

Measured against synthetic portfolios, not assumed.

- Tens of landlords: every page under about 120 ms of database time
- Hundreds: two problems, both measured and priced
- Functions ran in Washington, database in Frankfurt: moved, 647 → 338 ms
- 98 ms → 314 ms purely because another landlord's rows exist

Security

- Session in one HTTP-only cookie, against the library default
- 29 Row Level Security policies, proved by 134 tests
- Validation always runs on the server
- Service role key: one caller, unreachable from the browser

Still missing, and written down: rate limiting, MFA, an audit log.

With more time

Done Functions moved beside the database: 66% off the median page

1 Give two aggregate queries an indexable filter

2 More than one person on a portfolio

3 An audit log, and rate limiting on my own endpoints

In that order: measured effect over risk.

Thank you

Deployed. Tested at three levels. Every decision written down.

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